

Vishnu Kadiyala

Ph.D. Candidate, Computer Science · University of Oklahoma

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Research Focus

Multi-Agent Reinforcement Learning under partial observability; decentralized policy learning and implicit coordination via learned belief representations; learned communication for cooperative multi-agent systems; learning dynamics and optimization pathologies in cooperative MARL training; applications to autonomous driving and V2X.

Education

University of Oklahoma, Norman, OK Expected May 2027
Ph.D. in Computer Science

University of Oklahoma, Norman, OK May 2022
M.S. in Electrical and Computer Engineering
Thesis: Localization of Tables and Plots in Documents Using Deep Neural Networks

KLE Technological University, India May 2019
B.E. in Electronics and Communication Engineering

Technical Expertise

Machine Learning: Multi-Agent Reinforcement Learning, Deep Learning, Transformers, Attention Mechanisms, CNNs, U-Nets, Diffusion Models

Deep Learning Frameworks: PyTorch, JAX / Flax, TensorFlow, Keras

Simulation Environments: Highway-Env, MetaDrive, MPE (Speaker-Listener), SMAX, Overcooked

Programming: Python

Data & Compute: HPC / SLURM, CUDA, Multi-Machine Orchestration (Tailscale + SSH), Pandas, NumPy, Xarray

Tools: Git / GitHub, Linux, Reproducible ML Pipelines

Research Experience

Multi-Agent RL Research — University of Oklahoma 2025 – Present

- **Submitted to NeurIPS 2026:** *When Auxiliary Losses Fail: Non-Stationary Targets Induce Directional Gradient Noise*. Characterized a gradient-interference pathology in attention + auxiliary-loss architectures across both cooperative MARL and supervised learning. **290+ runs across 7 experiments**; three architectural fixes (stop-gradient on the auxiliary target, λ -annealing, mean-pool attention) recover lost performance.
- Cross-domain validation across Overcooked, Simple, and SMAX (MARL) and CIFAR-100 (vision), demonstrating the pathology is not RL-specific.
- Designed evaluation protocols that distinguish coordination from coincidence: ablations, counterfactual tests, and seed-pooled population estimators.
- Developing **AwareGate** (in progress, ICLR 2027 target): learned communication-gating policy combining cross-attention message fusion with a recurrent belief state, for cooperative multi-agent systems including V2X.

NSF AI2ES — ML Researcher ([GitHub](#)) 2023 – 2025

- Built a **Transformer-based** architecture with custom spatial / temporal embeddings for multi-modal irregular spatio-temporal data (ground stations + radar + satellite), achieving a **13× improvement** over the classical Marshall-Palmer baseline.
- Developed a **vision-based atmospheric visibility estimation** system using outdoor camera imagery for statewide inference beyond sparse ASOS sensor coverage.

- Designed and maintained reproducible ML training and evaluation pipelines on HPC infrastructure for large-scale multi-modal sensor datasets.
- Co-authored peer-reviewed paper: *Estimating Statewide Atmospheric Visibility From Camera Images* (AMS 2025).

NASA GeoCARB — Deep Learning Researcher ([GitHub](#))

2021 – 2023

- Designed **U-Net** architectures for methane hotspot detection from satellite imagery, achieving **95% accuracy**.
- Improved anomaly detection from 80% to **90.2%** using diffusion-based generative models on satellite observations.
- Built data-quality assessment frameworks for satellite observation datasets including preprocessing validation, augmentation strategies, and quality assurance pipelines.

Document Understanding with Deep Learning

Aug 2021 – May 2022

- Designed CNN-based architectures for **object detection and localization** of tables and plots in documents, achieving **99% detection accuracy**.
- Constructed annotated datasets with bounding-box labels and automated document generation pipelines for scalable training data.
- *Master's Thesis: Localization of Tables and Plots in Documents Using Deep Neural Networks.*

Selected Publications

Under Review

- **V. P. Kadiyala.** *When Auxiliary Losses Fail: Non-Stationary Targets Induce Directional Gradient Noise.* Under review at NeurIPS 2026.

Peer-Reviewed Conference Papers

- M. X. Sasser, M. Wilson Reyes, **V. P. Kadiyala**, A. Kurbanovas, K. J. Sulia, et al. *Estimating Statewide Atmospheric Visibility From Camera Images.* Proceedings of the 105th AMS Annual Meeting, 2025.
- E. Spicer, S. Crowell, F. Xu, **V. P. Kadiyala**, P. M. Klein, et al. *Exploring the Influence of Local Urban and Industrial Carbon-Based Pollutant Sources on Total Column Concentration Enhancements in Houston, Texas during TRACER.* Proceedings of the 104th AMS Annual Meeting, 2024.

Theses

- **V. P. Kadiyala.** *Localization of Tables and Plots in Documents Using Deep Neural Networks.* Master's Thesis, University of Oklahoma, 2022.

Teaching

Teaching Assistant — CS 2614 Computer Organization, University of Oklahoma

2024 – Present

- Lead weekly hands-on labs on digital logic, assembly-level programming, and hardware / software interface fundamentals.
- Develop lab documentation, debugging workflows, and troubleshooting guides; provide one-on-one mentoring across varied skill levels.

Industry & Engineering Experience

Test Automation Intern — Robert Bosch Engineering & Business Solutions

Jan 2019 – May

2019

- Developed hardware-in-the-loop (HIL) test automation pipelines for Engine Control Units (ECUs).
- Automated ECU software validation using ETAS LABCAR across hardware and digital fault layers.